

4.3 #s: 1, 7, 9, 16

4.4 #s: 7, 8

4.5 #s: 2, 5, 8, 9

Homework 8, Math 3210

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1: If f is a continuous function on $[-1, 1]$ which is differentiable on $(-1, 1)$ and satisfies $f(-1)=0$, $f(0)=0$, and $f(1)=1$, then show f' takes on the values 0, $\frac{1}{2}$, and 1 on $[-1, 1]$.

This is basically saying the slope of f on $(-1, 1)$ is 0, $\frac{1}{2}$ and 1 across the interval using the fact that $f(-1)=0$, $f(0)=0$ and $f(1)=1$.

By the MVT, since f is a continuous function across a closed interval $[-1, 1]$ and differentiable across the open interval $(-1, 1)$, there is at least one point c such that:

$$f'(c_1) = \frac{f(1) - f(-1)}{1 - (-1)} = \frac{1 - 0}{2} = \boxed{\frac{1}{2}}.$$

Likewise by MVT, if f is continuous across $[-1, 1]$, then it is continuous across $[-1, 0]$. Likewise if it is differentiable across $(-1, 1)$, then so it is differentiable across $(-1, 0)$.

$$f'(c_2) = \frac{f(0) - f(-1)}{0 - (-1)} = \frac{0 - 0}{0 + 1} = \boxed{0}.$$

Similar argument as above for the intervals $[0, 1]$, $(0, 1)$.

$$f'(c_3) = \frac{f(1) - f(0)}{1 - 0} = \frac{1 - 0}{1 - 0} = \boxed{1}.$$

So I have shown that f' takes on values $\frac{1}{2}$, 0, 1 across the interval $[-1, 1]$. $\square$

4.3 #s: 7, 9, 16

4.4 #s: 7, 8

5.1 #s: 2, 5, 8, 9

Homework 8, cont.

7. Prove that $\ln(x) \leq x-1$ for all $x > 0$. Hint: Analyze where $x-1-\ln(x)$ is increasing or decreasing.

Define a new function $g(x)$ to be $x-1-\ln(x)$. Then $g(x)$ is increasing on the interval where $g'(x)$ is greater than 0. Also, $g(x)$ is decreasing on the interval where $g'(x)$ is less than 0.

Consider any interval of the form $[r, 1]$ where $0 < r < 1$. Then we have a valid domain over which $g(x)$ is continuous and g is differentiable over $(r, 1)$. Now we can employ Theorem 4.3.5. So we find g' :

$g'(x) = 1 - \frac{1}{x}$. Since $r < 1$, $\lim_{x \rightarrow r^+} (1 - \frac{1}{x}) < 0$. This is true because $\frac{1}{r}$ is greater than 1. Likewise, $\lim_{x \rightarrow 1^-} (1 - \frac{1}{x}) \rightarrow 0$, but will never actually be 0, but always less than 0.

So, $g'(x) < 0$ across the entire domain of $(r, 1)$ for any $0 < r < 1$. By the Theorem, g is decreasing here.

Now we analyze the behavior over the interval $[1, r]$ where $r > 1$. So, $g'(x) = 1 - \frac{1}{x}$ and $\lim_{x \rightarrow 1^+} (1 - \frac{1}{x}) = 0$, but $g'(x)$ will always be positive because it never equals 0 and $1 - \frac{1}{x}$ will be a small positive number. Likewise, $\lim_{x \rightarrow r^-} (1 - \frac{1}{x})$ will be greater than 0 by the same logic. So, by Theorem 4.3.5 $g(x)$ is increasing on $(1, r)$.

Now, it suffices to say $g(x)$ can be re-written without impacting the truthfulness of the above. That is, $\ln(x) \leq x-1$ because $x-1-\ln(x)$ will always be positive.

4.3 #s: 9, 16

4.4 #s: 7, 8

5.1 #s: 2, 5, 8, 9

Homework 9, cont.

9. Prove Theorem 4.3.6: Let f be a continuous function on $[a, b]$ which is differentiable on (a, b) . Then f is non-decreasing on $[a, b]$ if and only if $f'(x) \geq 0$ for all $x \in (a, b)$, while f is non-increasing on $[a, b]$ if and only if $f'(x) \leq 0$ for all $x \in (a, b)$.

$\Rightarrow$ Assume f is continuous on $[a, b]$ and differentiable on (a, b) . If f is non-decreasing on $[a, b]$, then by definition $f(x_1) \leq f(x_2)$ where $x_2 > x_1$. Now perform the difference quotient between x_1 and x_2 . So, we get $\frac{f(x_2) - f(x_1)}{x_2 - x_1}$.

Since $f(x_1) \leq f(x_2)$, then $f(x_2) - f(x_1) \geq 0$.

Also, as given, $x_2 - x_1 > 0$. So we get a positive number for the difference quotient. Let $z \in [a, b]$. If $w > z$ then $w - z > 0$ implies $\frac{f(w) - f(z)}{w - z} \geq 0$, since $f(w) \geq f(z)$. Also, if $w < z$ then $w - z < 0$ implies $\frac{f(w) - f(z)}{w - z} \geq 0$ since $f(w) \leq f(z)$. So in either case, $f'(z) = \lim_{w \rightarrow z} \frac{f(w) - f(z)}{w - z} = \text{limit of something} \geq 0$. that is greater than 0

$\Leftarrow$ Assume $f'(z) \geq 0 \forall z \in [a, b]$. Now, suppose $x, y \in [a, b]$ where $x < y$. Since f is cont. on $[x, y]$ and differentiable on (a, b) , the MVT says $\exists c \in (x, y)$ s.t. $f'(c) = \frac{f(y) - f(x)}{y - x} \geq 0$. Since $f'(c) \geq 0$ and $y - x > 0$, $f(y) - f(x) \geq 0$ which is to say $f(x) \leq f(y)$. Then we have shown f is non-decreasing.

WLOG, it suffices to say the proof for non-increasing is similar but with reverse directions on the inequalities. $\square$

4.3 #16

4.4 #7, 8

5.1 #2, 5, 8, 9

Homework 8, cont.

16. Show that $\ln x$ is uniformly continuous on $[1, \infty)$ but not on $(0, 1]$

Using the definition of uniform continuity:

For all $\varepsilon > 0$ there exists a $\delta > 0$ s.t. for all $x, a \in D$,
 $|x - a| < \delta$ implies $|\ln(x) - \ln(a)| < \varepsilon$.

So, let $\varepsilon > 0$. Let $\delta = \frac{\varepsilon}{2}$. If $x, a \in [1, \infty)$ and $|x - a| < \delta$,
then $|\ln(x) - \ln(a)|$ by exercise 4.3.3 $\leq \left| \frac{x-a}{r} \right|$. Choose
 r to be $1/2$. Then $\left| \frac{x-a}{r} \right| = 2|x-a| < 2\delta = 2\left(\frac{\varepsilon}{2}\right) = \varepsilon$. So,
 $\ln x$ is uniformly continuous on $[1, \infty)$.

For the other interval, $(0, 1]$, we take note of the
result from exercise 3.3.7 which says any unbounded
function on a bounded interval cannot be uniformly continuous.
On that interval we have here $\lim_{x \rightarrow 0} \ln(x) = -\infty$. So,
 $\ln x$ is not uniformly continuous on $(0, 1]$,
which is what we wanted to prove. ■

Proof of 4.3.3:

By MVT, $\exists c \in [x, y]$ s.t. $g'(c) = \frac{\ln(y) - \ln(x)}{y-x} = \frac{1}{c}$. Since
we know $0 < x \leq c \leq y$, we can say $\frac{1}{y} \leq \frac{1}{c} \leq \frac{1}{x}$. So,

$$\frac{\ln y - \ln x}{y-x} \leq \frac{1}{x}; \quad \ln y - \ln x \leq \frac{y-x}{x}$$

The form used for #16 is $|\ln y - \ln x| \leq \left| \frac{y-x}{r} \right|$. This
is true because we are working across a positive domain.

4.4 HS: 7, 8

5.1 HS: 2, 5, 8, 9

Homework 8, cont.

7: $\lim_{x \rightarrow 0} x \ln x = \lim_{x \rightarrow 0} \frac{\ln x}{\frac{1}{x}} = \frac{-\infty}{\infty}$ which is indeterminate form.

So, using L'Hopital's Rule, $\frac{\frac{d(\ln x)}{dx}}{\frac{d(\frac{1}{x})}{dx}} = \frac{\frac{1}{x}}{-\frac{1}{x^2}} = -\frac{x^2}{x} = -x$

So, $\lim_{x \rightarrow 0} (-x) = 0$ ■

8: $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$. Plugging in 0, we get $\frac{0-0}{0} = \frac{0}{0}$

which is an indeterminate form. Use L'Hopital's rule.

$\frac{\cos x - 1}{3x^2}$. Then plug in 0 and we get $\frac{1-1}{0} = \frac{0}{0}$ which

is an indeterminate form. Use L'Hopital's rule again.

$-\frac{\sin x}{6x}$. Then plug in 0 and we get $\frac{0}{0}$ which is

an indeterminate form. Use L'Hopital's rule yet again.

$-\frac{\cos x}{6}$. Plug in 0 to get $-\frac{1}{6}$, which is the limit.

S.1 Hs: 2, 5, 8, 9

Homework 8, cont.

Prove that $\int_0^1 x dx$ exists by computing $U(f, P)$ and $L(f, P)$ for the function $f(x) = x$ and a partition P_n of $[0, 1]$ into equal sub-intervals. Then show that condition (5.1.7) of Theorem 5.1.8 is satisfied. Calculate the integral by taking the limit of the upper sums. Hint: Use Exercise 1.2.9.

$$U(f, P_n) = \sum_{k=1}^n M_k (x_k - x_{k-1}) = \sum_{k=1}^n \sup \{ f(x) : x \in [x_{k-1}, x_k] \} (x_k - x_{k-1})$$
$$L(f, P_n) = \sum_{k=1}^n m_k (x_k - x_{k-1}) = \sum_{k=1}^n \inf \{ f(x) : x \in [x_{k-1}, x_k] \} (x_k - x_{k-1})$$

For any partition of the interval $[0, 1]$, we get $P_n = (0, \frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots, \frac{n}{n} = 1)$.

So, for the function $f(x) = x$, the supremum of the function over some interval will be the larger of the two x -values. So, $f(x)$ will be $\frac{k}{n}$. Then each upper rectangle has an area of $(\frac{1}{n})(\frac{k}{n})$. Then to sum up each of those rectangles amounts to $\frac{2k}{n^2} = \frac{2}{n}$ be. Likewise for the lower we get $\frac{2k}{n^2} = \frac{2}{n}$.

So, $U(f, P_n) - L(f, P_n) = \frac{1}{n}$. $\lim_{n \rightarrow \infty} (\frac{1}{n}) \rightarrow 0$. So, condition 5.1.7 has been demonstrated.

Now, using the upper sum to calculate $\int_0^1 x dx$:

$$U(f, P_n) = \sum_{k=1}^n \left(\frac{k}{n}\right) \cdot \frac{1}{n} = \frac{1}{n^2} \sum_{k=1}^n k = \frac{1}{n^2} \frac{n(n+1)}{2} \text{ by 1.2.9.}$$

$$\text{So, } \frac{1}{2n} (n+1) = \frac{n}{2n} + \frac{1}{2n} = \frac{1}{2} + \frac{1}{2n}.$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2} + \frac{1}{2n}\right) \rightarrow \boxed{\frac{1}{2}}.$$

Homework 8, cont.

- 5: Let f be the function on $[0,1]$ which is 0 at every rational number and 1 at every irrational number. Is this function integrable on $[0,1]$? Prove that your answer is correct by using the definition of the integral.

(Lemma: There are infinitely many rational & irrational numbers between any two real numbers) Every Partition (length of the base of the rectangle will be real values. So, the supremum across any partition will be 1 and the infimum will be 0. So, $\int_0^1 f(x) dx = 0$, $\int_0^1 f(x) dx = 1$; the f is not integrable across $[0,1]$. ■

- 8: Suppose m and M are lower and upper bounds for f on $[a,b]$ in other words, $m \leq f(x) \leq M$ for all $x \in [a,b]$. Prove that $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$. What conclusion about $\int_a^b f(x) dx$ do you draw from this if the integral exists?

$$P = \{x_0 = a < x_1 < \dots < x_n = b\}; m_i = \inf \{f(x) : x \in [x_{i-1}, x_i]\} \geq m; M_i = \sup \{f(x) : x \in [x_{i-1}, x_i]\} \leq M.$$

$$U(f, P) = \sum_{i=1}^n M_i (x_i - x_{i-1}) \leq M \sum_{i=1}^n (x_i - x_{i-1}) = M(b-a)$$

$$L(f, P) = \sum_{i=1}^n m_i (x_i - x_{i-1}) = m \sum_{i=1}^n (x_i - x_{i-1}) \geq m(b-a)$$

So, $m(b-a) \leq L(f, P) \leq \int_a^b f(x) dx \leq \int_a^b f(x) dx \leq U(f, P) \leq M(b-a)$. Which is what we wanted to show. ■

S.1 #9

Homework 8, cont.

9. If k is a constant and $[a, b]$ is a bounded interval, prove that k is integrable on $[a, b]$ and

$$\int_a^b k dx = k(b-a).$$

Want to show $\int_a^b k dx = \int_a^b k dx = k(b-a).$

By previous question (#8), it follows that
 $m(b-a) = k(b-a) \leq \int_a^b k dx \leq \int_a^b k dx \leq k(b-a) = M(b-a).$

So, $k(b-a) = \int_a^b k dx = \int_a^b k dx = k(b-a).$

Which means the lower and upper integrals are equal and are $k(b-a).$

Also, $\int_a^b = \sup \{L(s, Q) : Q \text{ is a partition}\}$

$$\sum_{i=1}^n k(x_i - x_{i-1}) = k \sum_{i=1}^n (x_i - x_{i-1}) = k(b-a)$$

5.1 #5: 9

Homework 9, cont

9. If k is a constant and $[a, b]$ is a bounded interval, prove that k is integrable on $[a, b]$ and

$$\int_a^b k dx = k(b-a).$$

$$M_i = \inf \{ f(x) : x \in [x_{i-1}, x_i] \} = k$$

$$m_i = \sup \{ f(x) : x \in [x_{i-1}, x_i] \} = k$$

$$U(f, P) = \sum_{i=1}^n M_i (x_i - x_{i-1}) = k \sum_{i=1}^n x_i - x_{i-1} = k(b-a)$$

$$L(f, P) = \sum_{i=1}^n m_i (x_i - x_{i-1}) = k \sum_{i=1}^n x_i - x_{i-1} = k(b-a)$$

$$\int_a^b = \inf \{ U(f, P) : P \text{ a partition} \} = \inf \{ k(b-a) \} = k(b-a)$$

$$\int_a^b = \sup \{ L(f, P) : P \text{ a partition} \} = \sup \{ k(b-a) \} = k(b-a).$$

Since $\int_a^b k dx = \int_a^b k dx = \boxed{k(b-a)}$, so it is integrable on $[a, b]$ and it is $k(b-a)$.